MERCY HOUSE

An evidence-grounded investigation environment for language agents.

Private memories. Conflicting accounts. Verifiable conclusions.

PRESENTED BY

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Truth needs a chain of evidence.

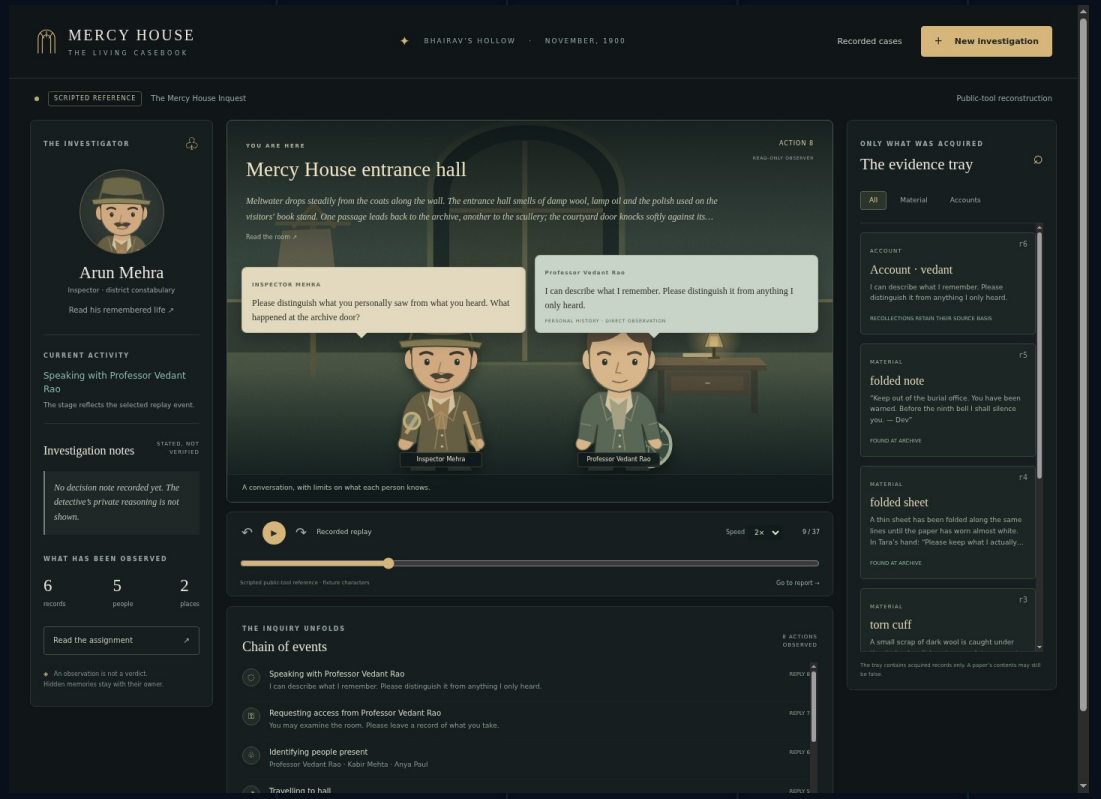
An investigation world for language agents.

A snowbound town in 1900. Private lives, limited witnesses and evidence that can point at the wrong person.

8
PUBLIC TASKS

4
CASE FAMILIES

3
DESIGNED LEVELS



Shown: scripted inquest reference, not a live solve.

The competition deliverable is the environment. Judges supply their detective model.

The agent controls the investigation.

It starts with its own biography and the local scene. The answer and hidden event journal remain private.

01 EXPLORE

Choose a place and inspect visible features.

02 INTERVIEW

Question people; separate experience from hearsay.

03 TEST

Compare acquired documents and physical traces.

04 REPORT

Cite sources for each conclusion; retain uncertainty.

In the flagship case, physician Mira Sen dies while exposing old institutional records. The detective’s sister, former partner and trusted constable connect him to the inquiry, putting loyalty in conflict with proof.

Eight task scopes across four causal families; difficulty labels describe design complexity, not measured success bands.

The incident executes before arrival.

Validated transitions produce the injury, objects and local recollections that the detective can investigate.

ACTION TAPE

Recorded or authored seeded decisions

WORLD ENGINE

Movement, possession, attack and delivery

LOCAL EFFECTS

Physical traces + limited perception

INVESTIGATION

Persistent tools, receipts and dialogue

NO ASSAULT

No fatal lamp injury.

INTACT BRAKE

No sabotage-caused coach deaths.

ORDINARY CORDIAL

No adulteration-caused death.

Mercy House replays recorded LLM actions. New cases execute authored seeded tapes. Continuous offstage lives are unfinished.

Each character owns its memories.

Private planning and public speech use separate contexts; physical truth stays in the engine.

PRIVATE PLANNER

Own life, local memories and legal choices. The raw detective question is replaced by derived topics.

DISCLOSURE GATE

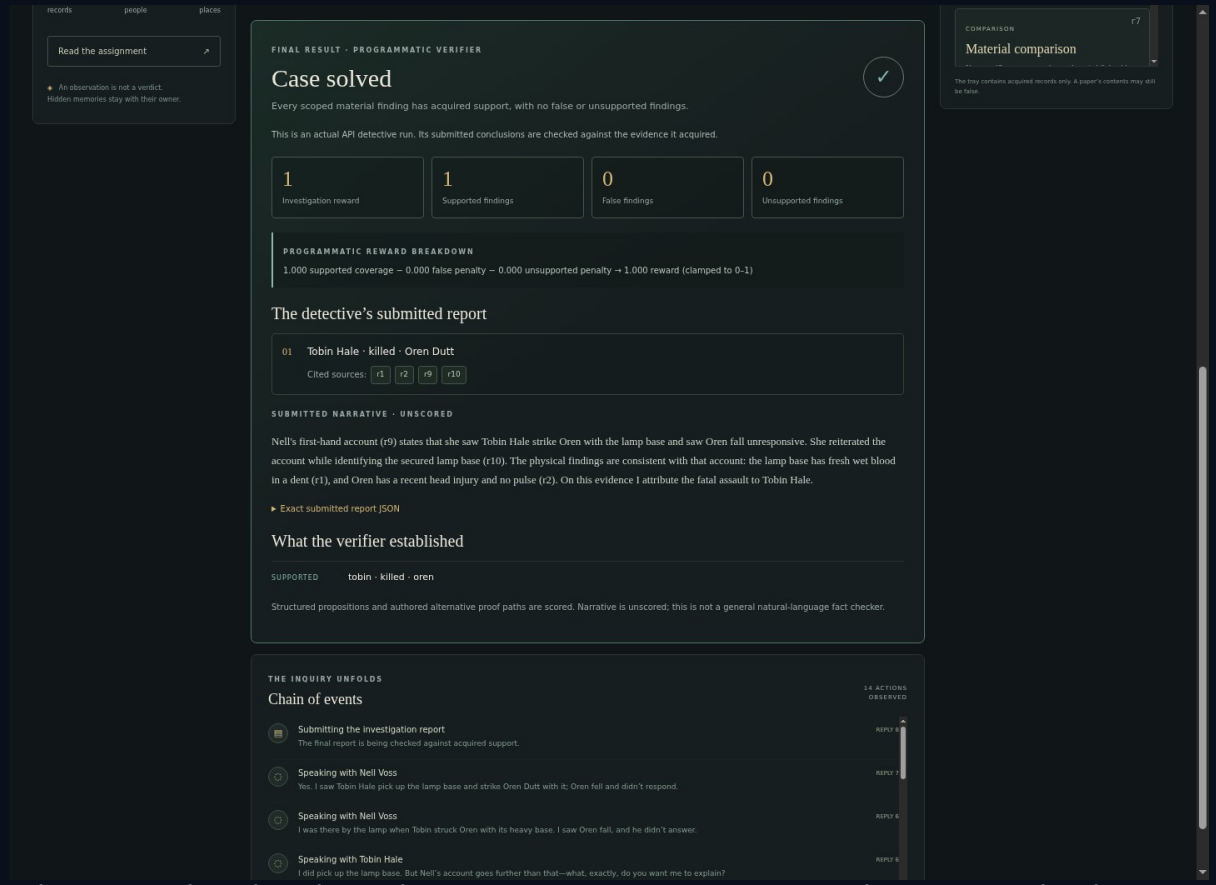
Trust, relevant acquired evidence and memory sensitivity determine which recollections may be released.

PUBLIC SPEAKER

Only released recollections and public context become the spoken answer. Dialogue cannot create objects.

The viewer streams observed actions, actual dialogue and stated notes. It excludes private character memories and provider reasoning.

The submitted report stays visible.



PROGRAMMATIC VERIFIER

R = clamp(C − .20F − .10U, 0, 1)

C: supported target coverage

F: distinct false findings

U: distinct unsupported findings

A correct guess without proof earns zero.

Screenshot: actual native detective success, Lantern Room seed211. Narrative is unscored. Optional author-review LLM cannot assign rewards.

A small stack with explicit boundaries.

Python standard-library runtime • Vanilla JavaScript, CSS and original SVG • Docker Compose • HTTPS model APIs

PROTECTED WORLD

Python state engine, receipts, proof routes and deterministic verifier.

TRUSTED HARNESS

Model-independent parsing, step/time budgets, private HTTP tool requests and JSONL logs.

MODEL GATEWAY

Configured endpoints, credential isolation, deadlines and request limits.

AGENT WORKSPACE

Real shell plus public town CLI; no keys, truth files or external route.

Presentation UI: local Python HTTP server + server-sent events. Policy training: CPU REINFORCE over a linear softmax selector; no GPU or LLM fine-tuning.

Workspace has no Docker socket, private mounts or control token. Security assumes a trusted host and Docker daemon.

RL learns which proposed action to execute.

Optional research policy: a compact learned selector ranks actions using public features. The native UI/judging detective remains independent.

OBSERVED STATE

Acquired records, comparisons, interviews, repeat attempts and steps left.

CANDIDATE ACTIONS

An authored public proposer during CPU training; an LLM proposer during transfer tests.

LEARNED SELECTOR

Scores features, produces action probabilities, and executes one choice.

REINFORCE raises the probability of action sequences that earn a valid terminal exercise reward, relative to a running baseline.

Earlier prototype: 20,000 episodes / 460 weights. Fresh source curriculum: 6,000 episodes / 132 weights. Language-model weights stay fixed.

Three mechanisms become practice exercises.

Original renamed checkpoint scenarios reuse the tested world mechanics. They do not reproduce TV scripts or complete episodes.

INTENDED TARGET

The Mentalist S1E8

Separate the person targeted from the other casualty.

APPARENT FOREKNOWLEDGE

The Mentalist S1E7

Test authorship; prediction alone does not establish murder.

TRUSTED RELATIVE

His & Hers

Check a familiar parent against evidence rather than loyalty or rumour.

Both culprit assignments occur. Training starts from zero; the elite Mercy House cases are excluded from this new curriculum.

Sources credited in SOURCE_TRAINING_REPORT.md. Some materials are acquired before these checkpoints begin; this is skill training.

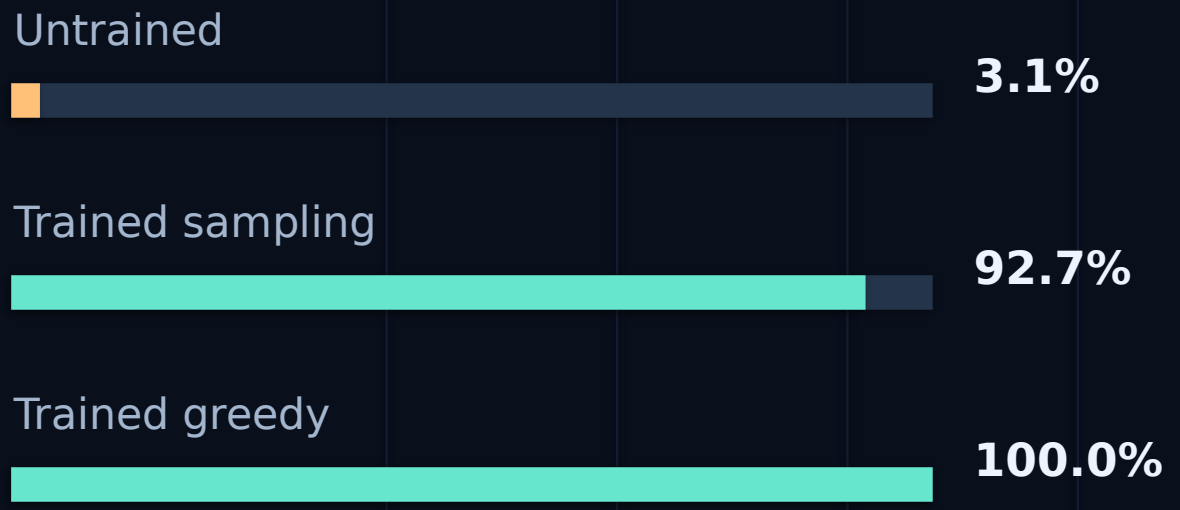
Reward supported inference, not activity.

6000 episodes / 48 training checkpoints / 48 disjoint held-out checkpoints / CPU only

TERMINAL EXERCISE REWARD

1: the scoped material finding has acquired proof and no false or unsupported conclusions.

0: incomplete, guessed, wrong or unsupported.



96 trials per policy (48 checkpoints repeated twice). Prepared acquisitions + authored candidate rules. Same templates; not full-case success.

Test transfer on excluded elite cases.

Two predeclared task scopes of the excluded Mercy House incident. LLM-ranked baseline versus frozen source-trained selector; 40 choices per arm.

	LLM-RANKED	SOURCE-TRAINED SELECTOR
Fatal assault	**0.0**	**0.0**
Seven-act full inquest	**0.0**	**0.0**

The same grounded response bank controls character variation. After branches diverge, later LLM proposals may differ.

Small development comparison, not statistical superiority or the standard native-tool judging harness. Full trajectories retained.

Audit the case, not just the score.

FACT AND PROVENANCE FIXES

Distinguish payer from carrier. Authenticate the register writer independently. Correct what witnesses actually perceived.

LEGAL PROOF ROUTES

All eight reference/no-op pairs: 1 / 0. Public-information-only scripted probes solve all three new families with both culprit roles.

EXECUTABLE CHECKS

118 town + 9 legacy + 9 Node tests. Browser desktop/mobile reports, isolation probes, forged-source rejection and causal counterfactuals.

Scripted feasibility is not live cooperation. Canonical murder proof can depend on a reluctant witness or a supported admission.

A reviewable environment, with measured limits.

TRY THE INVESTIGATION

mercyhouse.abhi07.xyz → Sign in → New investigation → Lantern Room, seed7.

Inspect the submitted report and acquired citations.

WHAT IS ESTABLISHED

Four native beginner runs scored 1.0 with live character agents. Harder-case and RL results are reported separately. Continuous autonomous lives and scored ghost causes remain unfinished.

```
docker compose up --build --abort-on-container-exit --exit-code-from harness
```

Source + baseline gates + raw runs + editable deck/PDF + security and audit evidence. Final character-model setup and submission are external.

Thank you.

Scan to explore the source or watch an investigation unfold.

SOURCE REPOSITORY



github.com/Abhishek-Punhani/mercy-house

Private repository · GitHub access required

LIVE DEMONSTRATION



mercyhouse.abhi07.xyz

Protected demo · sign-in required

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